Project: Worked examples
Title: Material strength margin
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

The factors of safety of a material against an applied (equivalent) stress on its yield and, when given, its ultimate strength, against a required factor; the strengths can be filled from a released material record so the margin cites its source.

WHAT IT CALCULATES	RELATION
Factor of safety against yield and against ultimate	$n_y = f_y/\sigma$; $n_u = f_u/\sigma$
Governing factor, the ultimate margin scaled to the yield basis by $k_u \geq 1$	$n_{gov} = \min(n_y, n_u/k_u)$
Utilisation against the required factor	$u = n_{req}/n_{gov}$

Basis: Design factor and factor of safety[1](§1-10).

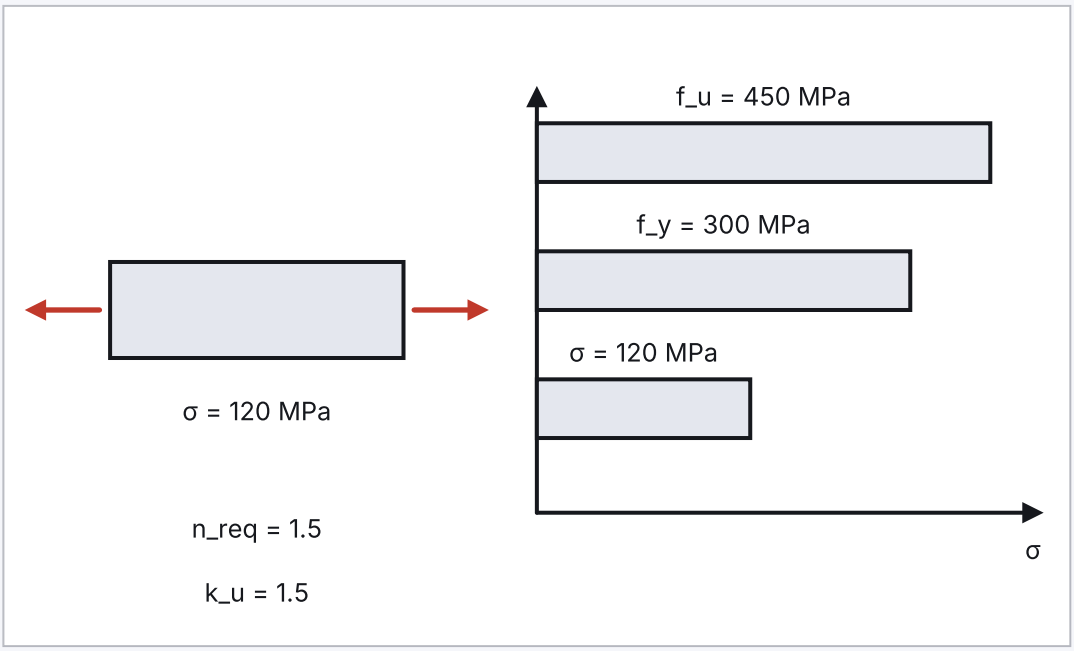


Figure 1 — A piece of the material under the applied stress, and the stresses as bars (drawn representatively, not compared). Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
sigma	Applied (equivalent) stress	120	MPa	spec: LOAD-SCHEDULE-01	0.01 … 5000 MPa
f_y	Yield strength	300	MPa	drawing: DRG-BR-701 rev A	1 … 5000 MPa
f_u	Ultimate tensile strength (optional)	450	MPa	drawing: DRG-BR-701 rev A	1 … 5000 MPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
k_u	Ratio of the ultimate to the yield safety factor required	1.5		spec: SPEC-MECH-03 design criteria	1 … 3
n_req	Required factor of safety on yield	1.5		spec: SPEC-MECH-03 design criteria	1 … 10

3. Assumptions

A1. Static loading; the applied stress is the equivalent stress at the critical point. (Fatigue and stability are separate checks.)

4. Working

```
[5] n_y = f_y / sigma                                          [1](§1-10)
= 300 MPa / 120 MPa
= 2.5
```

Factor of safety on yield

```
[7] n_u = f_u / sigma
= 450 MPa / 120 MPa
= 3.75
```

Factor of safety on ultimate strength

```
[8] n_gov = min(n_y, n_u / k_u)
= min(2.5, 3.75 / 1.5)
= 2.5
```

Governing factor (ultimate margin scaled to the yield basis)

```
[9] u = n_req / n_gov
= 1.5 / 2.5
= 0.6
```

Required over governing factor of safety

5. Results

SYMBOL	VALUE	DESCRIPTION
n_u	3.75	Factor on ultimate
n_y	2.5	Factor on yield
n_gov	2.5	Governing factor
u	0.6	Utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Factor of safety against the required factor	u = 0.6	u ≤ 1	0.6	[1](§1-10)

7. Applicability

No warnings.

8. Verification

```
Template mech.material-margin@1.0.1 Content hash 7ddefccebe71d022 Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.